

FLYING AD-HOC NETWORKS (FANETS)

RLFR ALGORITHM IN UAVNETSIM

NET
ROUTING



FANET ROUTING CHALLENGES & MOTIVATION

01

The Networking Problem

- High 3D UAV mobility (10 m/s) causes rapid wireless link degradation.
- Greedy geographic forwarding gets trapped in **routing voids**.
- Standard Q-routing explores blindly, triggering massive latency spikes (>90 ms).

02

The RLFR Solution

- Bounded Latency: Enforces a strict 40 ms QoS delay budget via a **Risk Table**.
- Energy Efficiency: Selects dynamic relay power levels (25, 50, 75, 100 mW).
- Cooperative Mesh: Shares state values in beacons for fast convergence without central servers.

The Three Core Pillars of RLFR

1. Multi-Objective Utility

Rather than optimizing purely for distance or delay, RLFR formulates a joint utility combining delivery success, packet latency, and transmission power.

$$U = K - c_1\tau - c_2W$$

2. Safe Latency Exploration

Maintains an explicit Risk Table $R(s, a)$. In the Boltzmann action policy, choices that historically breach the 40 ms latency deadline are strongly penalized.

$$\pi(s, a) \propto \exp(Q - c \cdot R)$$

3. Distributed Bellman Updates

Drones piggyback their local state-value $V(s)$ into periodic 0.5 s HELLO beacons. Neighbors learn cooperatively with zero central coordination.

$$Q \leftarrow (1-\alpha)Q + \alpha[u + \lambda \max Q + u \Sigma V_j]$$

How RLFR Routes Packets Hop-by-Hop



Observed State Variable	Symbol	Network Role
Candidate Drone Battery	b	Avoids dying relays
Received SNR & Link Gain	ξ	Ensures high-quality wireless channel
Accumulated Hop Count	q	Prevents excessive path stretching
One-Hop Queuing Delay	t	Detects local buffer congestion
Active Neighbor Count	ϖ	Measures local swarm density

Loop Elimination: Cache Ω

Every drone maintains a cache Ω of `(source_id, packet_id)` tuples. If a packet loops back due to mobile topology shifts, it is dropped instantly.

Result: Zero broadcast storms and elimination of routing cycles.

Mathematical Formulation: Utility & Power Control

Multi-Objective Utility (Equation 3)

$$U^{(k)} = \kappa - c_1 \cdot \tau - c_2 \cdot w$$

Quantifies the multi-hop transmission reward when forwarding data packet k .

Relay Power Action Space

$$a = (\text{next_hop}, p), \quad p \in \{25, 50, 75, 100\} \text{ mW}$$

By choosing power dynamically based on link SNR ξ , drones conserve battery and shrink the spatial interference footprint.

Parameter	Value	Physical Meaning
Delivery Success (κ)	1 or 0	Confirmed via destination ACK
Normalized Latency (τ)	$c_1 = 0.8$	End-to-end packet delay / $T_{\max}$
Transmission Energy (w)	$c_2 = 0.6$	$w = p \cdot (z / r)$ (Power $\times$ Duration)

The Single-Objective Failure

Greedy forwarding picks maximum-distance hops with high bit error rates. RLFR balances distance against transmit energy and delay penalties.

Mathematical Formulation: Safe Exploration & Risk

Safe Boltzmann Policy (Equation 2)

$$\pi(s, a) = \exp(Q(s, a) - c \cdot R(s, a)) / \sum \exp(\dots)$$

The risk weight $c = 0.5$ penalizes actions that violate latency constraints.

Risk Table Update (Equations 5 & 6)

$$\begin{aligned} \tau^{(k)} &= 1(\tau > \mu) \quad [\mu = 40 \text{ ms threshold}] \\ R(s, a) &\leftarrow (1 - \beta) \cdot R(s, a) + \beta \cdot \tau^{(k)} \end{aligned}$$

$\beta = 0.8$ enables rapid exponential smoothing to detect fading channels instantly.

Why Safe Exploration is Essential

- **No Blind Exploration:** Unlike ϵ -greedy, RLFR never explores links that risk breaching the 40 ms deadline.
- **Real-Time Adaptation:** If a drone moves behind an obstacle, its $R(s, a)$ jumps to 1.0, and the policy immediately switches to an alternate path.
- **Guaranteed QoS:** Mission-critical video and telemetry stay within delay tolerances.

Distributed Bellman Update with Shared Experience

Distributed Bellman Equation (Equation 4)

$$Q(s, a) \leftarrow (1 - \alpha) \cdot Q(s, a) + \alpha [u + \lambda \cdot \max_{\hat{a}} Q(s', \hat{a}) + v \cdot \sum_j V_j(s_j)]$$

Hyperparameters: Learning rate $\alpha = 0.70$, Discount factor $\lambda = 0.90$, Shared experience weight $v = 0.05$.

Beacon Value Piggybacking

Every UAV calculates its local value function $V_j(s_j) = \max_a Q(s_j, a)$ and embeds it into periodic 0.5 s HELLO beacons.

Zero extra protocol messages; fully decentralized.

Fast Cooperative Convergence

Instead of waiting for ACKs from the destination, a relay immediately learns from its neighbors' evaluated state values via $v \cdot \sum_j V_j$.

Code Implementation Architecture in UavNetSim

Package: `routing/rlfr/`

- `rlfr_packet.py`:
 - `RLFRHelloPacket`: Beacons coordinates, battery, and $V(s)$.
 - `RLFRackPacket`: Carries κ , delay τ , and link SNR ξ .
- `rlfr_table.py`:
 - Implements Q-table, Risk-table $R(s,a)$, and loop cache Ω .
 - Computes safe Boltzmann exploration policy.
- `rlfr.py`:
 - Protocol controller: packet queuing, CSMA/CA scheduling, and Bellman updates.

Simulator Integration

- `entities/drone.py`: Registered in protocol factory map.
- `routing/parameters.py`: Configured paper hyperparameters.
- `main.py` CLI: Executable via `--routing RLFR`.
- 3D Web UI: Interactive selection at `localhost:8000`.

Clean Interoperability

Shares the exact same SimPy environment, CSMA/CA MAC, and physical channel models as the other 9 protocols for zero-bias benchmarking.

Final Test Performance of All 10 Protocols

Protocol	Delivery Ratio (%)	E2E Delay (ms)	Average Hops	Throughput (Kbps)	MAC Delay (ms)
Greedy	91.47%	100.16 ms	1.47	1413.1	26.70 ms
DSDV	97.01%	15.59 ms	1.52	1366.4	5.99 ms
GRAD	90.83%	7.42 ms	1.00	1444.2	0.00 ms
OPAR	59.91%	8.79 ms	1.34	1417.8	13.82 ms
QRouting	82.94%	17.36 ms	2.38	1012.7	11.54 ms
QFANET	91.26%	91.74 ms	2.31	919.7	9.39 ms
QGeo	88.27%	22.22 ms	2.71	821.7	14.15 ms
QMR	78.89%	15.56 ms	2.45	875.9	12.83 ms
Baseline_DRL	91.47%	100.16 ms	1.47	1413.1	26.70 ms
RLFR (10th)	81.45%	16.47 ms	2.45	891.4	7.45 ms

Best delay among RL models: [RLFR - 16.47 ms]
Latency reduction vs Q-FANET: [$> 5.5\times$ faster] | Vs Baseline DRL: [$> 6.0\times$ faster]

Identical Conditions: 10 UAVs · Gauss-Markov 3D · CSMA/CA · Seed 2025

A large number of drones are flying in a coordinated pattern over a coastal landscape at sunset. The drones are connected by a network of glowing lines, suggesting a swarm or a networked system. The background shows a coastline with a winding road, a body of water, and mountains in the distance. The sky is filled with soft, golden light from the setting sun.

THANK YOU

Questions & Discussion :)